

Soumyajyoti Dutta

(979) 326-2896 | soumyajyotidutta23@gmail.com | <https://xoumyax.github.io/> | [linkedin.com/in/soumyajyotidutta](https://www.linkedin.com/in/soumyajyotidutta)
Empirical LLM Research | Post-Training & Evaluation | AI for Cyberdefense & Adversarial ML

EXPERTISE

Core Competencies:

- **LLM post-training:** supervised fine-tuning and curriculum-staged training to align foundation models (135M–~9B, incl. sparse MoE) to intended behavior — e.g. teaching an agent to clarify rather than hallucinate on ambiguous intent.
- **Evaluation & Diagnosis:** task-specific metrics (lexical + syntactic + SMT-based semantic comparison), LLM-as-judge and user-simulation harnesses used as training-time evaluation signals; frontier-API baselines and stage-wise ablations that trace *where* a method's gains stop and *why*.
- **AI × security:** LLM systems for automated cyberdefense (threat-intel rule generation, IOC/TTP extraction), adversarial ML (built a competition's top-ranked adversarial attack suite), and ML malware detection at 2M+-sample scale.
- **Efficient training & inference:** single-GPU & multi-node SLURM training with FFT and PEFT (LoRA/QLoRA), MoE, grammar-derived hybrid tokenization, and vLLM inference with multi-turn KV reuse.
- **Research → public output:** USENIX Security '27 Cycle-1 submission (second targeting Cycle 2), ACSAC 2026 acceptance with an open-source PyPI release, Springer chapter (2022), and a deployed full-stack applied-LLM platform.

Technical Stack:

Python (PyTorch, HuggingFace Transformers, TRL, TensorFlow, NumPy, Pandas, scikit-learn, Matplotlib), SQL/SQLite, DeepSpeed, FSDP, DDP/NCCL, CUDA, SLURM, vLLM, Z3/SMT, Docker, AWS, Git, CI/CD, FastAPI/Flask/Django, React, C++, Rust

PROFESSIONAL EXPERIENCE

Texas A&M University — Botacin's Lab

Research Assistant / Machine Learning Engineer (Applied AI)

College Station, TX

Spring 2024 – Present

- Built end-to-end LLM post-training pipelines: data curation, SFT / curriculum training, strategic evaluation, and deployment across ~a dozen model families (135M–~9B) on multi-node SLURM/GPU clusters.
- **YaraGen**, sponsored by **Laboratory For Physical Sciences, UMD & Booz Allen Hamilton** — grammar-oriented minimal LLMs for automated YARA rule generation (**submitted, USENIX Security '27, Cycle 1**):
 - * Lifted a **139M-parameter BART** generator to **92% semantic correctness** (93.7% in-window) on 22.7K held-out real-world rules — matching a fully fine-tuned 8B model (93.8%) with **57× fewer parameters** and 17–20× the generation throughput.
 - * Benchmarked frontier APIs on structured generation: fully fine-tuned Llama and Ministral 8B models beat few-shot GPT-5.4 (84.0%) by **4–8 pp** and led zero-shot GPT-5.4 (73.5%) and Sonnet 5 (75.0%) further; the 139M pipeline leads few-shot GPT-5.4 by ~10 pp, and frontier parse rates of 97–99.8% show the gap is semantic, not syntactic.
 - * Ran **150+ post-training runs** across BART, CodeT5p, T5Gemma, SmolLM2, LFM2.5 (230M–1.2B), Gemma 3 (270M–1B), Llama 3.1 8B / 3.2 1B, and Ministral 3 8B, spanning FFT, LoRA/QLoRA, and seq2seq vs. causal ablations.
 - * Designed the **YaraAST hybrid tokenizer** — grammar-derived atomic tokens for rule structure over base subwords for IOC content — yielding +4 pp accuracy, 11% per-rule context reduction, and 5% faster training (3.8% fewer FLOPs on 16 GPUs).
 - * Co-designed a **55M BiLSTM NER Scooper** that masks IOCs behind placeholders (99.4% precision / 99.6% recall), adding up to +10 pp correctness and decoupling IOC size from the context window.
 - * Trained **LLMScooper**, a LoRA adapter (r=32, ~85M params) that turns the 8B generator's own backbone into an IOC tagger at **0.999 span F1**, swapped on GPU-resident weights.
 - * Assembled three corpora (305K crawled in-the-wild rules; 15M synthetic from 459 curated templates; 15M grammar-fuzzed) and built *Brownie & Puff* composite evaluators (BLEU + parse/compile + **Z3 SMT semantic comparison**) as training-time signals; only the composite correlated positively with test accuracy (BLEU Spearman down to -0.70).
 - * Downstream on EMBER malware clusters, generated rules reach **96±4% TP at 0.8% FP** (99.7% generation success) vs. AutoYara's 93±6% at 0.5% FP; released the framework, agentic web chatbot, and models as open-source artifacts.
 - * *In progress:* extending YaraGen with tool use, verification and repair, designing verified multi-task SFT and an RLVR pilot with executable, formally checked rewards.

- **IOCScooper++** — IOC/TTP extraction from real threat reports (**targeting USENIX Security '27, Cycle 2**):
 - * Trained **ScooperNER**, a 50.6M-parameter extractor (MLM pretraining → fine-tuning, 36 IOC types): **0.894 micro / 0.925 macro** span-exact F1 on held-out vendor and government reports, beating fine-tuned encoder baselines at 3–7× its size, with 0.950 F1 on explicit MITRE ATT&CK IDs.
 - * End to end on 47 held-out reports, extraction feeds frozen YaraGen at 0.86/0.88 precision/recall with **100% rule validity** at **0.97 IOC coverage** across 344 generated rules.
 - * Extending to grounded (entity, action, IOC) tuples mapped open-set onto all 637 ATT&CK techniques via procedure-example retrieval, with a co-annotated benchmark of real malware reports.

- **AutoPYara** — shipped an open-source rule-generation package (PyPI; **ACSAC 2026, accepted**): +14 pp coverage, 7.1 pp lower variance, and +10% threat-hunting accuracy over AutoYara (prior state of the art).

Cognizant Technology Solutions

Kolkata, India

Junior Software Engineer — Client-Embedded Consulting (Client: TJX Companies)

Aug 2021 — Aug 2022

- Client-embedded consultant for TJX Companies (Fortune 500): shipped a logistics-automation platform; optimized Python/REST services (latency, concurrency, CI/CD) and led cross-functional code reviews.

Software Engineering Intern — converted to full-time offer

Feb 2021 — Jun 2021

- Built web features for a client logistics platform; resolved support tickets via ServiceNow triage and Splunk tracing.

SELECTED PROJECTS

Ambiguity-Aware Research Agent | *Co-authored (manuscript in prep.) — led model development & training* 2026

- Aligned a deep-research agent to ask a clarifying question on underspecified intent instead of answering the wrong question, via four-stage SFT of Gemma-4-E4B (sparse MoE, ~9B/4B) on one 80GB GPU with vLLM.
- Built a three-axis evaluation with GPT-4o user simulation and LLM-as-judge; lifted pause-detection F1 from 0.168 to 0.573 and showed clarified answers still trail the no-pause baseline, isolating where SFT alone falls short.

Depixelation Recoverability Frontier | *Personal vision-security research — in progress* 2026

- Designing an open-source benchmark of when pixelated text is recoverable: a **360-cell** difficulty grid with a **10-seed** protocol that recasts depixelation as constrained search with an exact render-and-verify check, plus retrieval-augmented recognition with calibrated abstention on 4×H100.

Sturdy Fishstick — Full-Stack Applied-LLM Platform | *FastAPI, React, local LLMs via Ollama* 2025

- Deployed a full-stack platform that has scored ~750 listings from 10+ sources, using categorical scoring that makes 1–2B local models reliable judges and FTS5-grounded RAG that rules out hallucinated counts.

Multimodal & Vision Evaluation | *ECEN 642 — Image Processing & Computer Vision* 2024

- Evaluated a DeepSeek vision-language model against human-annotated ground truth; earlier: carotid-artery ultrasound CNN classification (Springer chapter, 2022) and WildFire spatiotemporal prediction.

ML Malware Detection & Adversarial Attack Suite | *1st Place; still #1 after five course offerings* 2023

- Built an end-to-end Windows PE malware classifier (EMBER/BODMAS/Benign-NET, 2M+ samples; LightGBM/XGBoost) — 96.2% accuracy, 0.97 F1 — served from a Docker HTTP endpoint; designed the competition's best-performing adversarial attack suite.

EDUCATION

Ph.D. in Computer Science — Texas A&M University Spring 2024 – Expected Spring 2028

- *Research focus:* LLM post-training, evaluation, and interpretability for automated cyberdefense.

M.S. in Computer Engineering — Texas A&M University Fall 2022 – Fall 2023

- *Coursework:* NLP, Deep Learning, Computer Vision, ML Cyberdefenses, ML Theory, Data Mining.

B.Tech. in Electronics & Communication Engineering — SRM University Fall 2017 – Spring 2021

PUBLICATIONS & MANUSCRIPTS

YaraGen — minimal grammar-oriented LLMs for YARA rule generation USENIX Security '27, submitted (Cycle 1)

IOCScooper++ — grounded IOC/TTP extraction from threat reports USENIX Security '27, in prep. (Cycle 2)

AutoPYara — automated rule generation; open-sourced on PyPI ACSAC 2026, accepted

Proactive Deep Research Agents — ambiguity-aware clarification (co-authored) 2026, in preparation

Carotid-Artery Ultrasound Classification — CNN-based medical imaging Springer book chapter, 2022

TEACHING & OUTREACH

Teaching Assistant, CSCE 313 — Introduction to Computer Systems, Texas A&M Fall 2026

- Lead lab sections and grading; write hands-on Linux assignments on processes, IPC, threads, and synchronization.

Teaching Assistant, CSCE 439/704 — Data Analytics for Cybersecurity, Texas A&M Spring 2026

- Taught clustering, supervised ML, and anomaly detection to 37 students; mentored 9 team projects.

Partner & Presenter, TAMU CS Day — interactive AI/cybersecurity demos for high-school students 2024, 2025